# Mori Seiki NLX 2500 Operator Manual — Cleveland Site Copy

CNC turning center. Controller Fanuc 0i-TF. Installed in Auto-Turn-1 cell, Plant 1 (Cleveland). Two units: NLX-08 (primary) and NLX-09 (backup).


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```


## 1. Machine identification

| Field | Value |
| --- | --- |
| Site machine IDs | `NLX-08`, `NLX-09` |
| Make / model | Mori Seiki NLX 2500 (CNC turning center) |
| Controller | Fanuc 0i-TF — see `PDM-MAN-FANUC-0iTF-001` |
| Chuck | 10-inch 3-jaw hydraulic |
| Spindle max | 4,000 rpm, 22 kW |
| Turning length × diameter | 705 mm × 366 mm |
| Primary use | Turning and boring Part `34521-A` (AISI 4140 hydraulic actuator housing) for Vortex Motors |

## 2. Operating envelope

| Parameter | OEM machine limit |
|---|---|
| Max spindle speed (chuck) | 4,000 rpm |
| Max feed rate | 30,000 mm/min rapid / 10,000 mm/min cutting |
| Max bar capacity | Ø 80 mm through-spindle |
| Coolant pressure | 15 bar through-tool |

> **Cutting parameters for Part 34521-A OP30** (rough turn, finish turn, bore) are defined in work instruction `PLM-WI-34521A-OP30-RevD`. Stay within the OEM envelope above.

## 3. Boring operation guidance (Part 34521-A OP30)

The Ø 38.000 +0.025 / −0.000 mm bore on Part 34521-A is the most critical feature produced on this machine. See control plan `QMS-CP-34521A-RevC` for the inspection regime and PFMEA `QMS-FMEA-34521A-RevB` for failure-mode context.

- Use the dedicated boring bar set BB-34521A-01 (carbide insert TNMG 16-04 grade KC5010).
- Coolant must be ON for the entire bore pass (heat buildup causes thermal drift and is the most common bore-OOT cause per PFMEA failure mode FM-007).
- Do not override feed below 80% — chatter at lower feeds will degrade surface finish.
- Insert change at 200 pieces or when bore-gauge check trends > +0.015 mm from nominal, whichever is first.

## 4. Daily startup checklist

1. Power up controller; reference X and Z axes.
2. Run program `NLX_WARMUP_10MIN`.
3. Check coolant level and concentration.
4. Verify chuck pressure: 35 bar nominal.
5. Confirm tool offsets against setup sheet.

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**Document control / sign-off**

| Role | Name | Signature | Date |
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| Document Owner | R. Bertolino, Process Engineering — Cleveland | (on file) | 2024-04-08 |
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